Lecture Transcript by Slide

Week 2 · Appendix C: 12-Lead Electrocardiography · 26 min

Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

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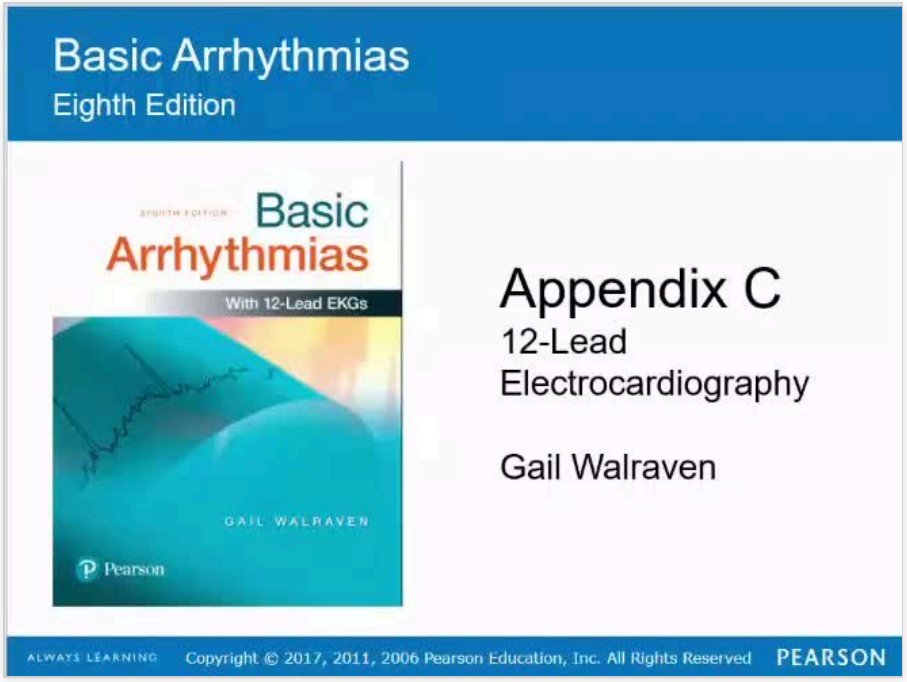
0:19:47 Horizontal Plane Leads

0:20:07 Standard Layout of 12-Lead EKG

0:24:11 Limitations of EKGs

1 Appendix C: 12-Lead Electrocardiography (title slide)

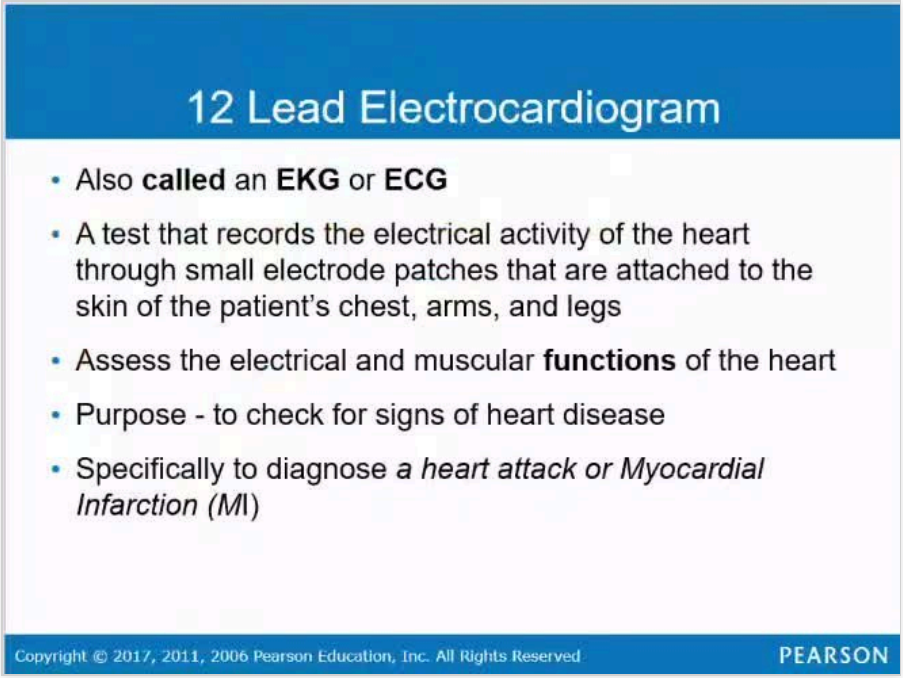
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0:00:00 part of week 2's lecture is going to be about 12 lead EKGs. So also known as an EKG or an ECG,

2 12 Lead Electrocardiogram

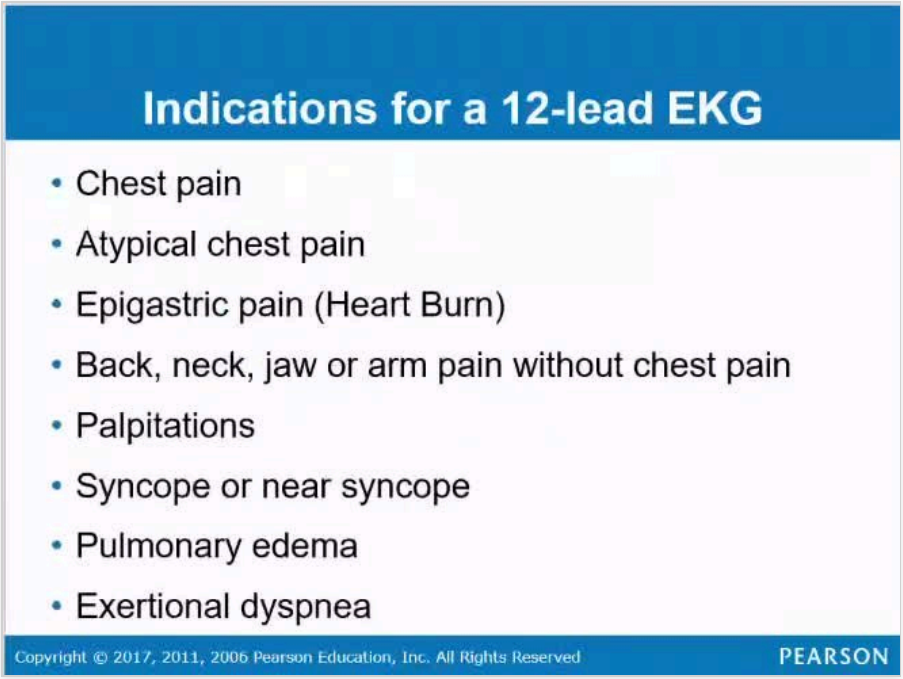
0:00:10 – 0:01:03



0:00:14 this is a test that records the electrical activity of the heart through electrode patches that are attached to the patient's skin on the chest, arms, and legs. The reason why we do a 12 lead EKG is we would like to assess the electrical and muscular functions of the heart to check for signs of heart disease. And most specifically the biggest indication for performing a 12 lead EKG is to diagnose somebody that's having a heart attack or the appropriate name for it is myocardial infarction or MI. So you should be able to associate an MI with someone having a heart attack. So what are the specific indications

3 Indications for a 12-lead EKG

0:01:03 – 0:02:42

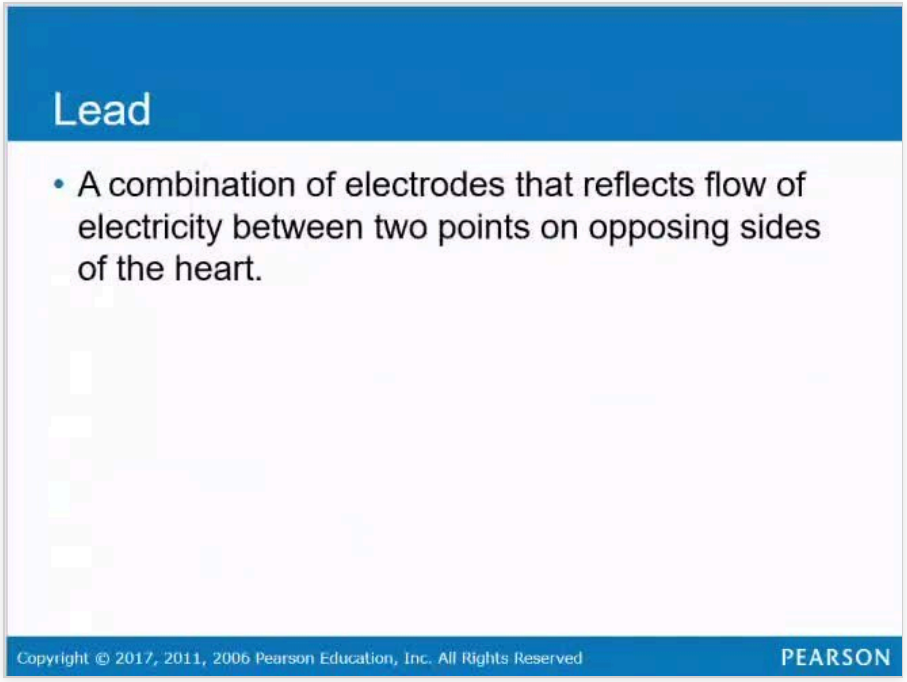


0:01:07 or the symptoms that a patient may present with while having a heart attack? So somebody that's having an MI or a heart attack might be complaining of chest pain, they might be complaining of atypical chest pain, or it may mimic something that we call heartburn. So they might have pain or like a bubble or something in the middle of their chest and sometimes it's really hard to differentiate whether the patients have a heartburn or they're actually having chest pain from an MI.

0:01:39 Sometimes this pain can occur in their back, neck, it can be jaw pain, it can be arm pain without chest pain or it can be radiating pain that goes down the leg or the back. They also may be complaining of what they feel like is a fluttering or palpitations in their chest, they may get to the point where they feel like they're going to pass out, otherwise known as syncope. If the myocardial infarction is bad enough that it affects the function of the left ventricle, they may go into pulmonary edema where the fluid, the heart, is injured from the heart attack. It can't push blood forward so fluid backs up into the lungs and they have what's called pulmonary edema. These patients may also be complaining of shortness of breath with movement, walking, any kind of exertion. So we talk about

4 Lead

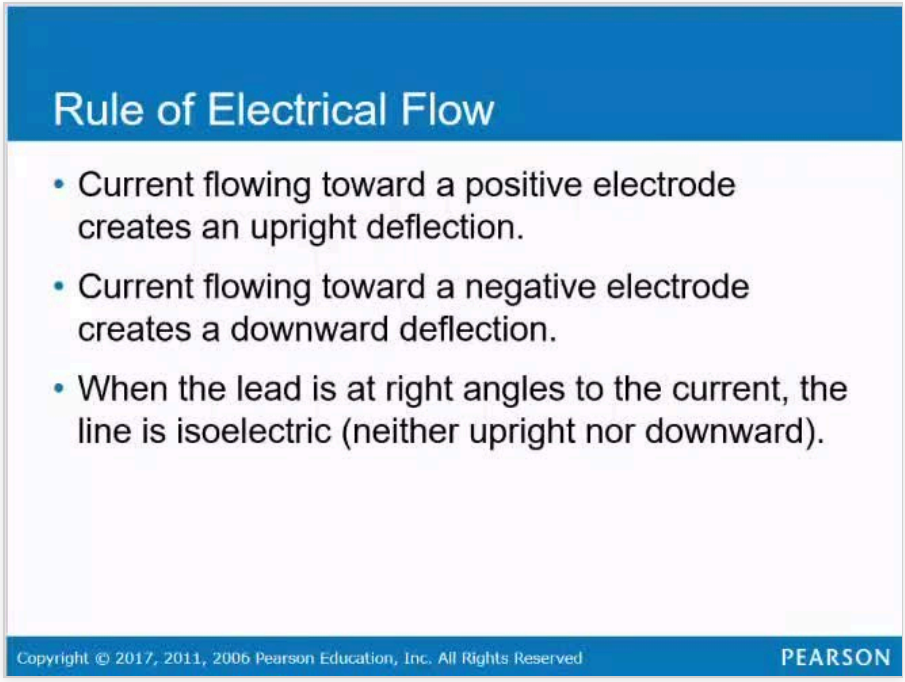
0:02:42 – 0:03:22



0:02:44 putting patches on the patient's chest and arms and legs. We put these patches there because what we want to do is create what's called a lead. A lead is a combination of electrodes. So we would be looking at two electrodes and from one electrode to the next electrode that what gets recorded will be the flow of electrical activity between those two opposing points. So that's what a lead is and some of these leads are neutral and some of

5 Rule of Electrical Flow

0:03:22 – 0:04:40

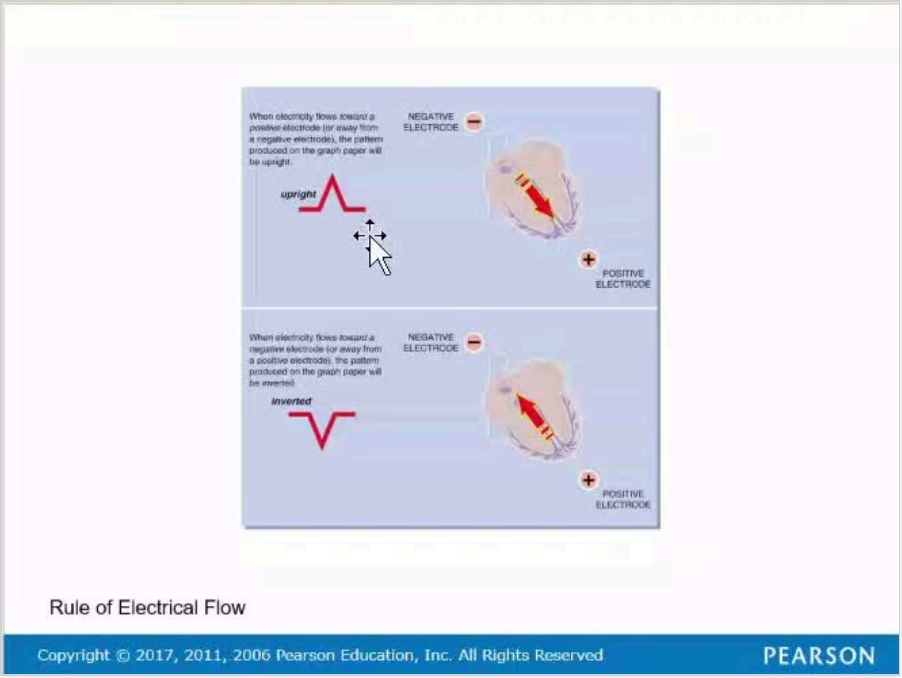


0:03:29 these leads are negative and some of these leads are positive. So rule of electrical flow here, electrical flow between two leads. So if electrical flow from the heart is flowing towards a positive electrode it's going to create a positive deflection. It's flowing towards a negative electrode. It will create what we call a downward or negative deflection. When the lead is at right angle to the current, the line is isoelectric, so it's neither positive or negative. The isoelectric line is the neutral line in the middle of the EKG. So when we see electrical flow going through a straight line across the middle, that's the isoelectric line. It is not positive, it is not negative. When we see a pattern formed above it, it's positive.

0:04:35 When we see a pattern formed below it, it's negative. And this is what I'm

6 Rule of Electrical Flow (diagram)

0:04:40 – 0:05:32

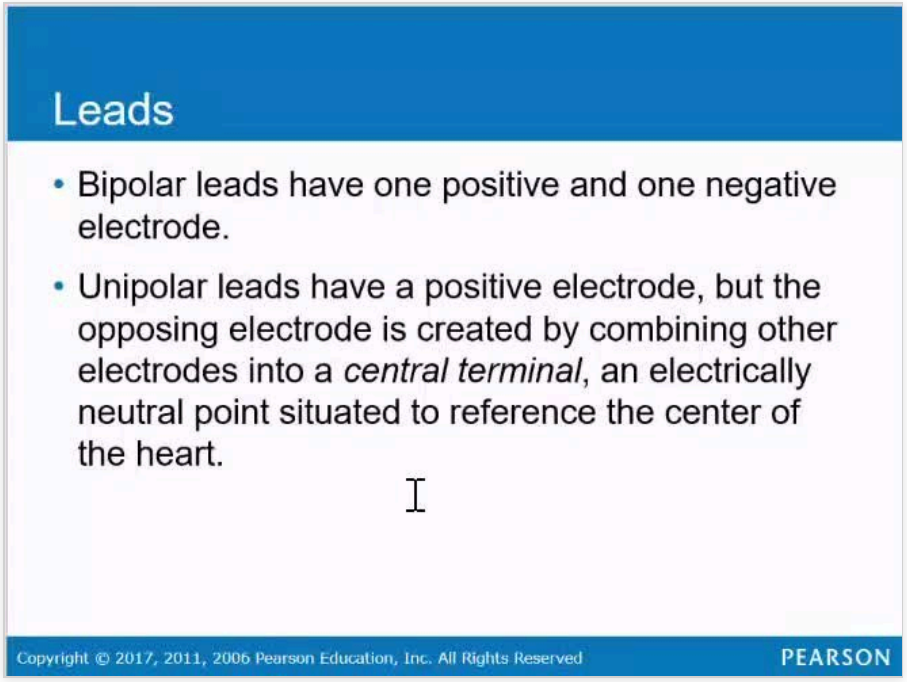


0:04:49 we have electrical flow going down towards the ventricles, which would cause ventricular depolarization, and the lead down here is positive. So we have electrical flow going towards that positive electrode. We're going to get an upright or a positive deflection.

0:05:12 This lead up here on the right arm is a negative electrode. So if we get electrical flow going towards the right arm and causing depolarization, we're going to get a negative deflection.

7 Leads: Bipolar and Unipolar

0:05:32 – 0:06:39

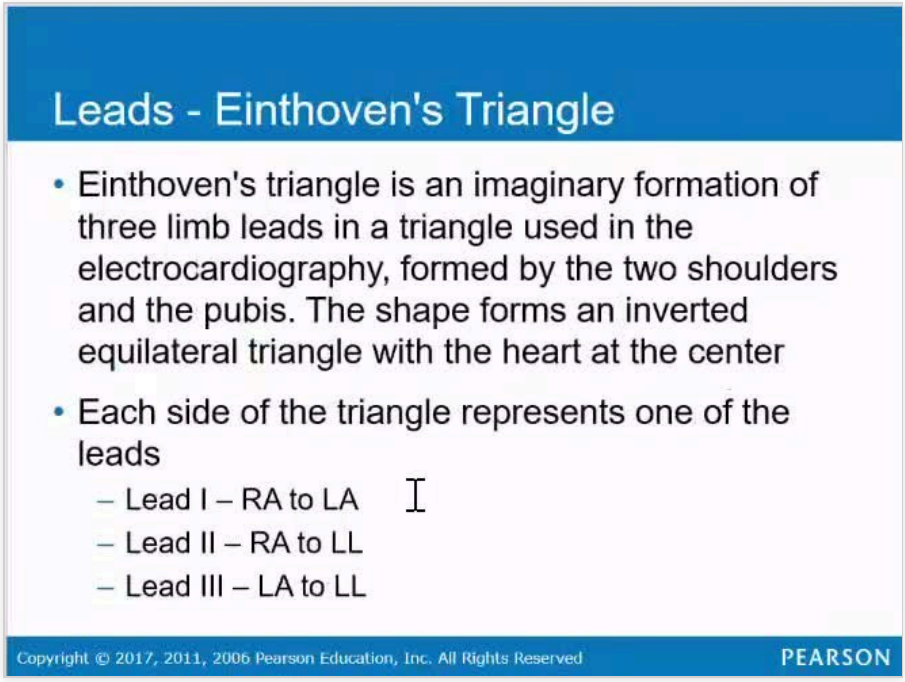


0:05:34 Most of these leads that we look at on the chest are what we call bipolar leads. So bipolar leads have one lead that is positive and one lead that is negative. We do have some exceptions to that rules some leads that we call unipolar. Unipolar leads have a positive electrode, but the other end of the electrode is created by combining other electrodes into what we call a central terminal,

0:06:09 an electrically neutral point situated to a reference right in the center of the heart. I'll explain a little bit more about these. These are the augmented leads, but what happens is we have a number of leads that create a central terminal, and then the unipolar leads go from the central terminal out towards a specific electrode, like the left arm or the left leg.

8 Leads - Einthoven's Triangle

0:06:39 – 0:07:56

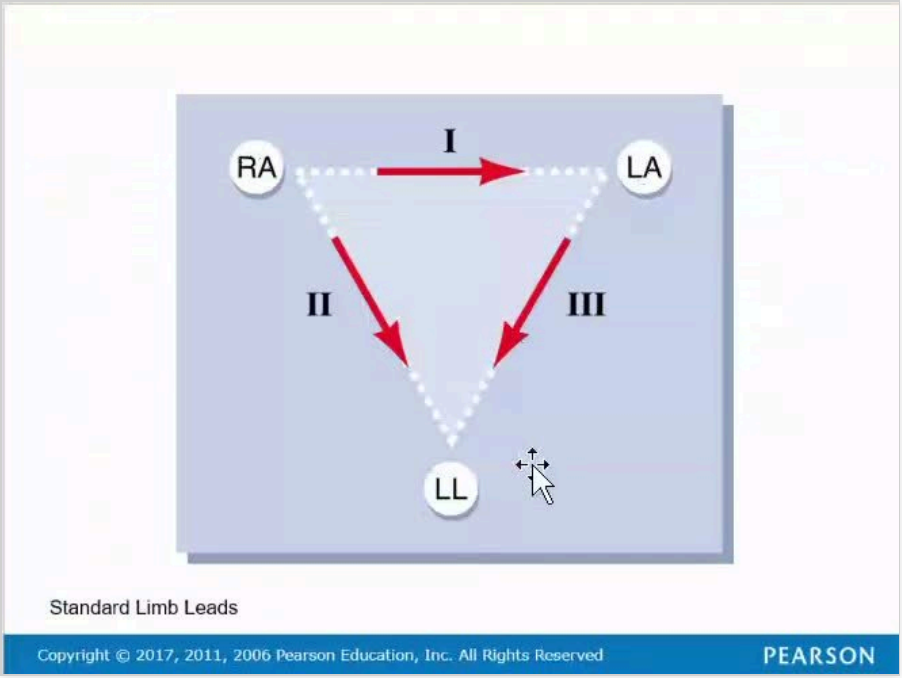


0:06:41 So if we go way back and discover the origin of how we created these leads, we would have to go back to Eindfthoven's triangle. So Eindfthoven's triangle is an imaginary formation of three limb leads in a triangle used in monitoring or recording electrocardiographic representation of the heart's electrical activity. So these three leads are formed by two shoulders in the pubis, So we're gonna have three leads. One going to all three of those spots or electrodes. The shape of this triangle is an inverted equilateral triangle with the heart at the center. So we're measuring electrical activity with the heart in the center. So lead one goes from the right arm to the left arm.

0:07:43 Lead two goes from the right arm to the left leg and lead three goes from the left arm to the left leg. And this is what this triangle

9 Standard Limb Leads (diagram)

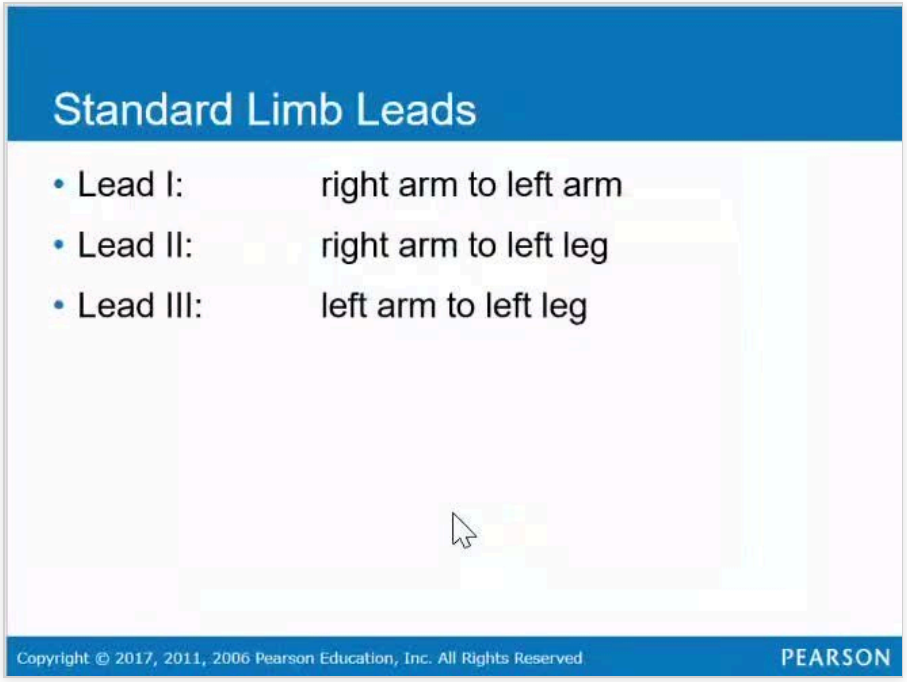
0:07:56 – 0:08:32



0:07:58 looks like. So we have right arm, left arm, left leg with the heart sitting in the middle of this triangle. So lead one is going to go across the top of the chest from the right arm to the left arm. Lead 2 is going to go across the center of the chest from the right arm down to the left leg. And then lead 3 is going to cover the lateral portion of the heart. It's going to go from the left arm down to the left leg. And then this next

10 Standard Limb Leads

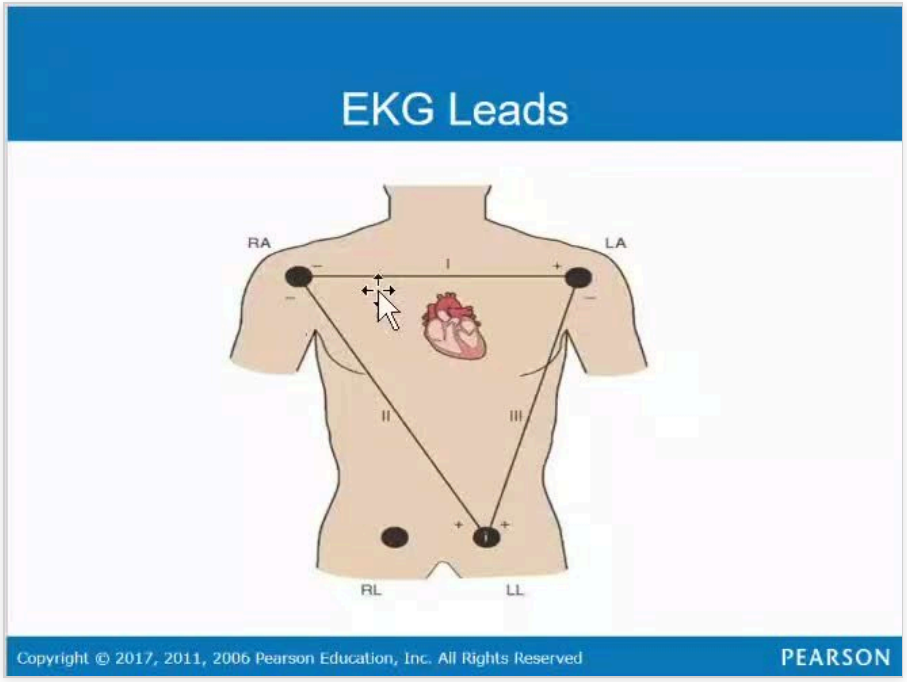
0:08:32 – 0:08:52



0:08:32 slide is just repeating that information. Lead 1, lead 2, lead 3. You should understand where the where the electrodes are placed and what direction of the electrical flow each of these leads records and then here we have the triangle on

11 EKG Leads

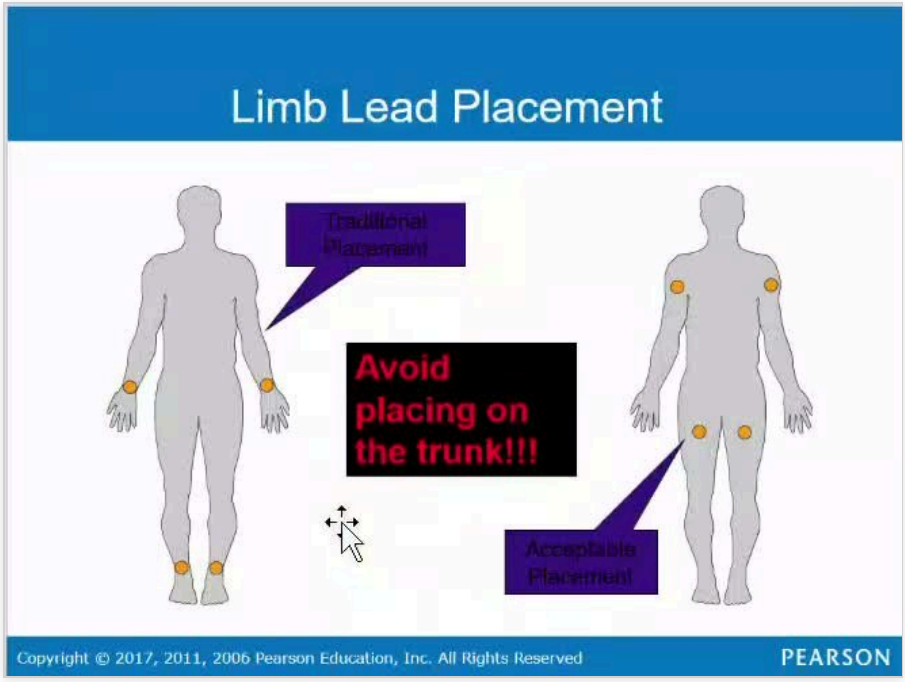
0:08:52 – 0:10:07



0:08:56 a real person right arm left arm left leg And we have each three of those leads now the other thing that I would like to point out here is Over by the right arm lead or electrode, sorry you can see that this right arm is negatively charged. So this is a negative electrode. So when we have flow going back towards the right arm on the monitor or on a recorded piece of paper for an EKG, this is going to create negative deflections below the isoelectric line. The left arm electrode is a positive electrode and the left leg is a positive electrode. So anytime we have flow going towards the left leg or towards the left arm, we are going to get positive deflections on the monitor or the EKG recording. So now that you understand a little bit more about leads and placements, we're going to talk more specifically about 12 lead EKGs because that's ultimately the purpose of this

12 Limb Lead Placement

0:10:07 – 0:11:23

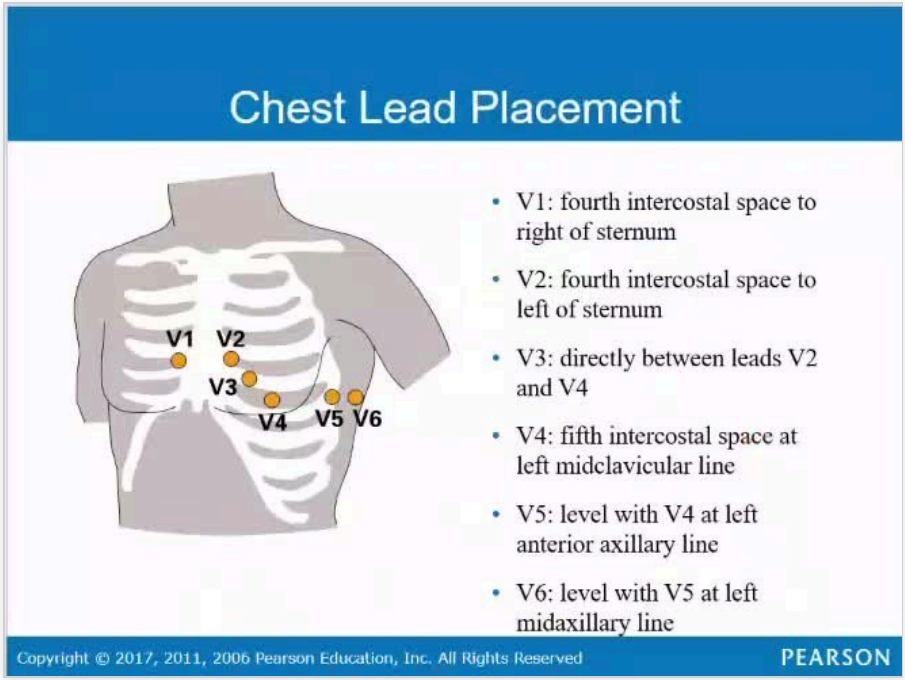


0:10:09 lecture. So when we're performing a 12 lead EKG, we don't put the electrodes on the right arm, I'm sorry, the right shoulder, the left shoulder. The leads actually should be placed down on the right wrist, the left wrist, the left ankle, and the right ankle. So these are traditional

0:10:35 acceptable limb placements for a 12-lead EKG. Sometimes we can't always place them there based on, you know, if somebody's had an amputation or somebody has like a cast from a broken leg or something like that. So we may actually have to modify these placements and go more towards what you are capable of doing with the patient with certain conditions. So sometimes we may have to move these electrodes up towards the shoulder or up towards the base of the legs. Whenever possible though we should be placing the leads on the wrists and ankle. Okay then the other leads that we need to place are what's

13 Chest Lead Placement

0:11:23 – 0:14:27



0:11:29 called the pre-cordial leads. These are the chest leads. And so B1 is a sticker or an electrode that we are going to place on the fourth intercostal space just to the right of the sternum. So what I mean by an intercostal space is the space between the ribs. So the first rib is up here. So this would be the first intercostal space, the second intercostal space, the third intercostal space down to the fourth intercostal space. And this is the sternum, so we're going to place that just to the patient's right of the sternum. Not your right, the patient's right. V2 is going to go in the fourth intercostal space to the left side of the sternum. Then the next electrode we are going to place is going to be V4. So we're going to skip V3 for a second. We're going to move down to V4 and V4 is going to go in the fifth intercostal space left mid clavicular line. Now clavicular is referring to the clavicle or the collarbone,

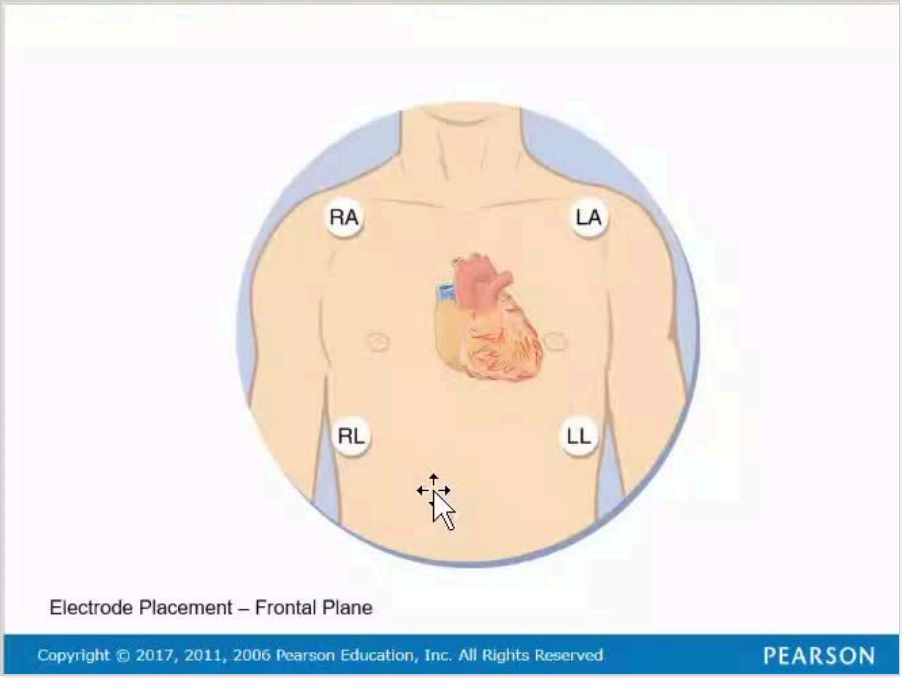
0:12:43 which is this collarbone up here that you guys can all feel that sticks out a little bit. So we want to go right in the middle of that bone right there and then we want to go down to the fifth intercostal space and place electrode V4. And we'll go back and now we're going to place the third, the V3 electrode, and that is going to go halfway in between V2 and V4.

0:13:14 And we're going to move on and place the fifth pre-cordial lead, which is going to be the fifth intercostal space, the anterior axillary lining. So axillary is referring to armpit. So right at the front portion of the armpit, we're going to go straight down to the fifth intercostal space and place V5. And then we're going to place our last electrode, which is pre-cordial lead 6. This is also going to be in the fifth intercostal space, but this is going to be mid axillary line. So if we

lift the patient's arm up right in the middle of the armpit, straight down to the fifth intercostal space, that's where pecordial lead 6 is going to be placed. So we're placing actually 10 leads on the patient's chest, right? The right wrist, the left wrist, the left ankle, the right ankle, and then V1 through V6. And these leads are going to create 12 electrical views of the heart. And we'll get into talking about that shortly. So

14 Electrode Placement - Frontal Plane

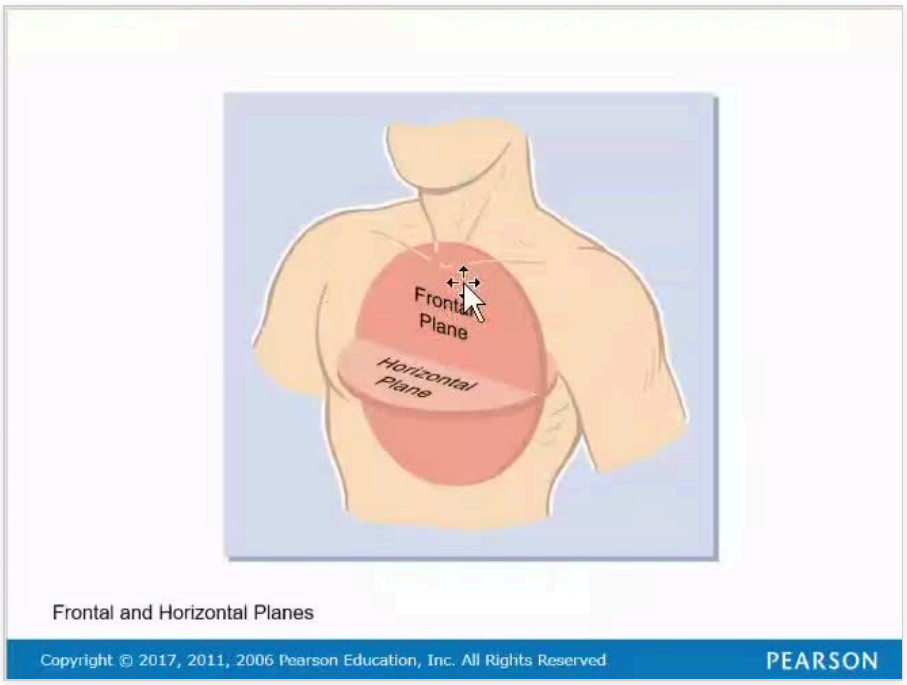
0:14:27 – 0:14:46



0:14:29 the limb leads that we place are actually going to assess what we call the frontal plane of the heart. So vertical up and down is going to be the frontal plane.

15 Frontal and Horizontal Planes

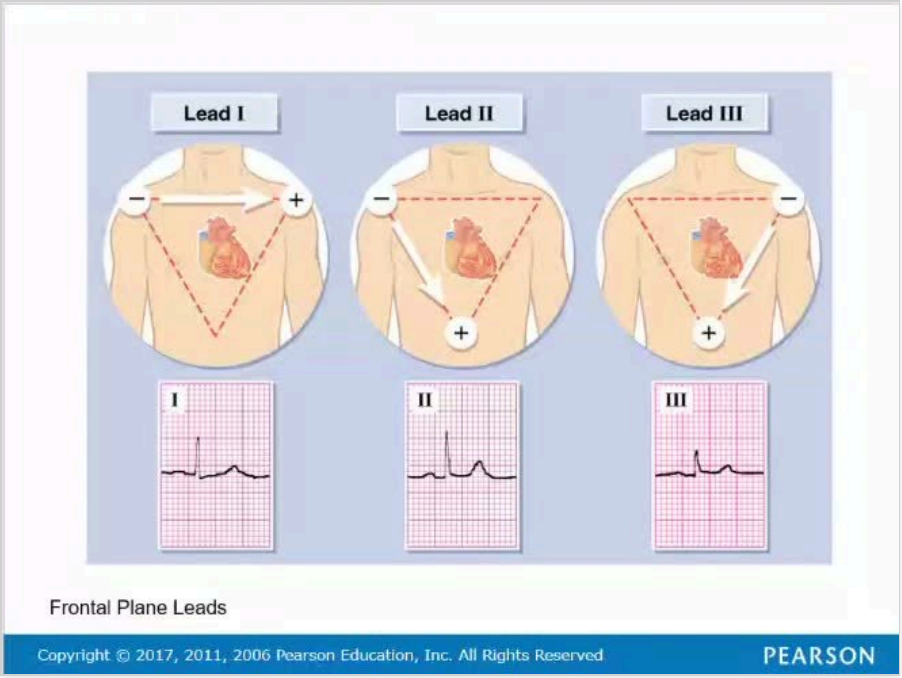
0:14:46 – 0:15:16



0:14:46 So if I go to the next slide, the limb leads are going to be placed and it is going to assess the vertical plane or the frontal plane of the heart. When we place the precordial leads on the chest going around the heart, then those leads are going to assess the horizontal plane.

16 Frontal Plane Leads

0:15:16 – 0:16:44

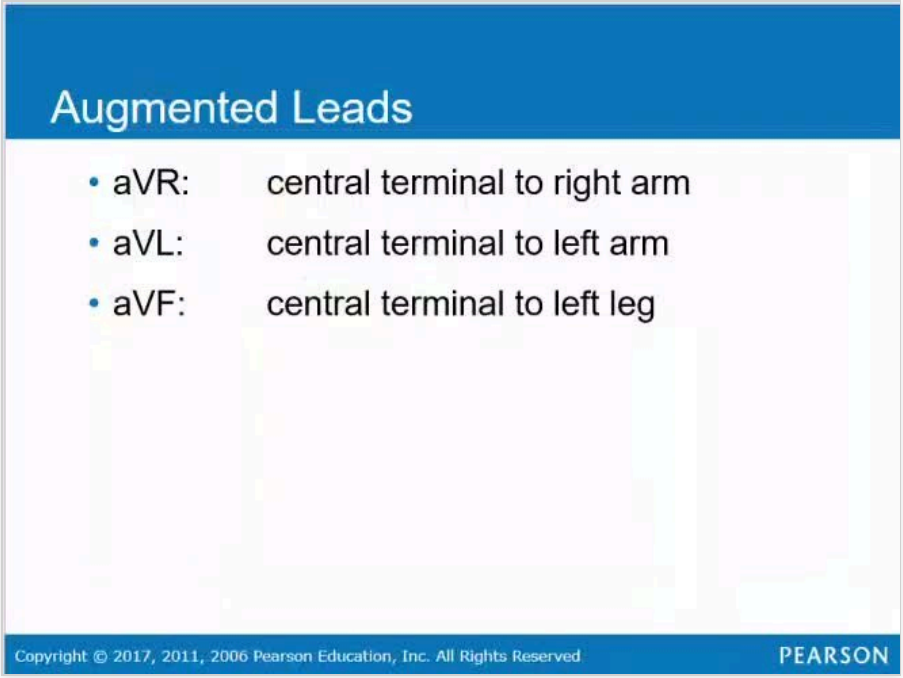


0:15:18 So as we move on here, these are going to start showing us the specific recordings on paper for each one of these leads. So lead one, remember, is the right arm to the left arm. It is going cover some of the anterior surface of the heart and the superior surface of the heart. And so it is flowing towards a positive electrode. So what we're going to see here is positive recordings.

0:15:53 We're going to get into what the waveforms stand for shortly. Lead two, which is the standard monitoring lead, gives us from the right arm to the left leg. Again this is flowing towards a positive electrode. So we see atrial depolarization in the P wave, which is positive, the QRS, which is positive, that's ventricular depolarization, and then the T wave, which is ventricular repolarization are all positive in that lead. And then we go to lead three. We're going to go from the left arm to the left leg. The left leg again is a positive electrode, so all of our waveform recordings there are also going to be positive. And before I was talking

17 Augmented Leads

0:16:44 – 0:17:45

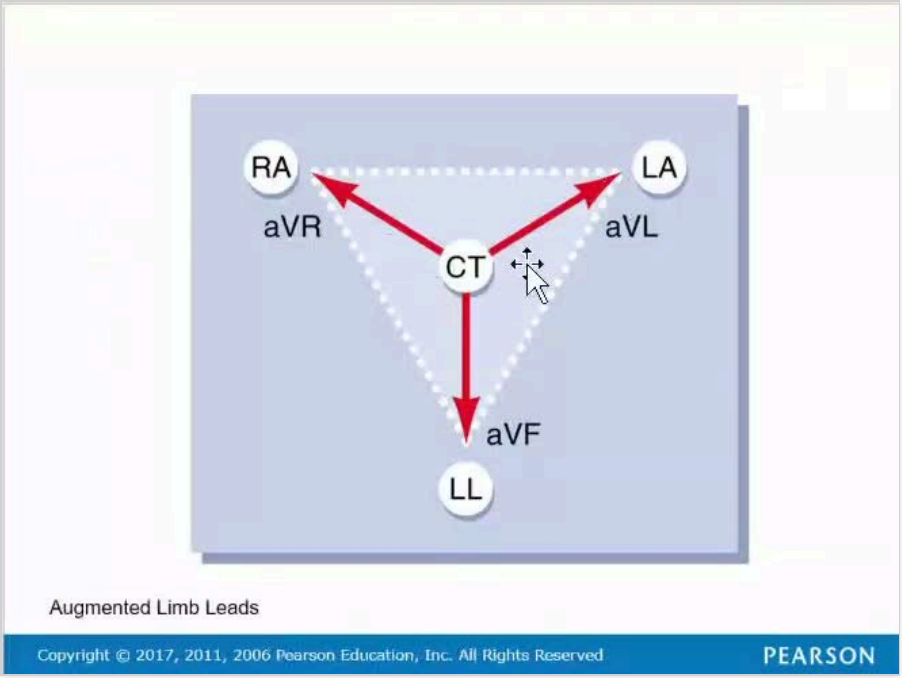


0:16:47 about unipolar leads that a combination of leads creates a central terminal and then we go from the central terminal to a specific electrode which creates a lead. So we take this central terminal which is going to sit right in the middle of the heart to the right arm and that is going to be classified or named as AVR, central terminal to the right arm because it's an augmented lead.

0:17:22 Central terminal to the left arm is going to be AVL, again an augmented lead or a central terminal to the left arm, and then central terminal to the left leg is going to be AVF. And so we're going get an electrical recording on the 12 lead EKG for all three of these leads. And so this is what

18 Augmented Limb Leads

0:17:45 – 0:18:17

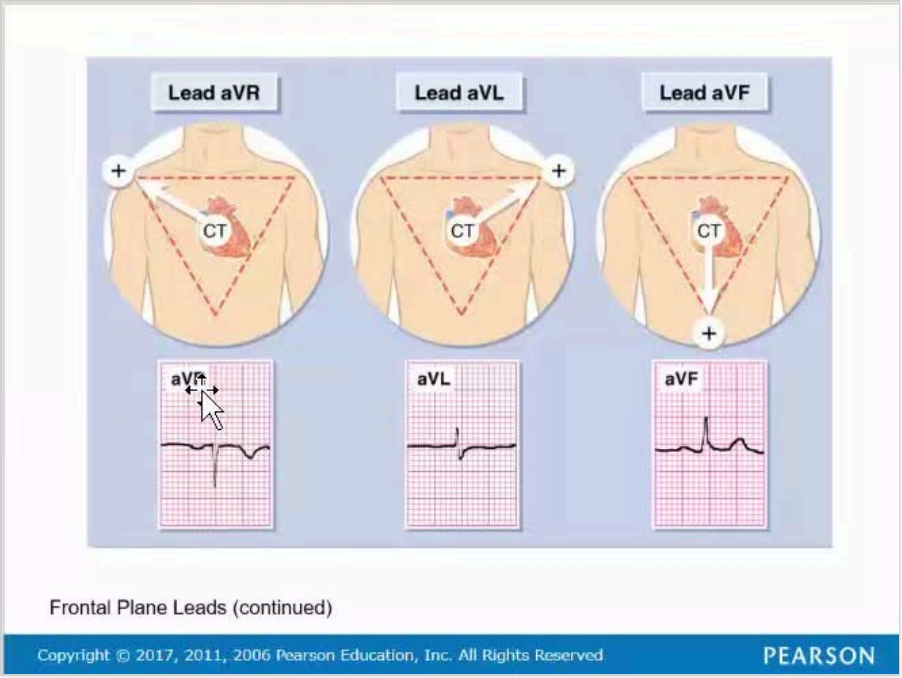


0:17:46 it looks like. The heart, this is Eindhoven's triangle in the middle. The heart's going to sit in the middle. The leads create what we call a central terminal. Central terminal to the right arm AVR. Central terminal to the left arm AVL. Central terminal to the left leg AVF.

0:18:05 So these are what are known as an Augmented Leads, Augmented Limb Leads. Those are the Unipolar Leads.

19 Frontal Plane Leads (continued)

0:18:17 – 0:19:31

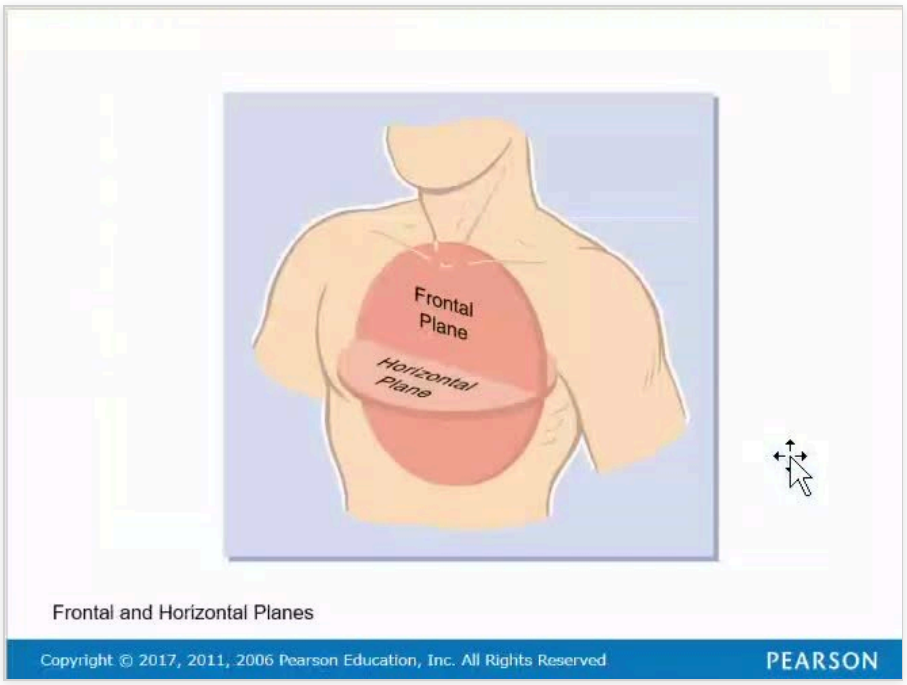


0:18:17 Now we're going to move on to what we see on a 12 lead EKG recording for each one of these leads. So remember central terminal to the right arm. The right arm is a negative electrode. Okay, so we're going to see all of the waveforms are going to appear below the isoelectric line or negatively record it. So the P wave is inverted upside down because it's going towards a negative electrode, the QRS and the T wave are all going to be negative flowing towards a negative electrode.

0:19:00 AVL, central terminal to the left arm is going to be, it's a positive electrode but it gives us 50-50 in terms of positively depicted waveforms, negatively depicted waveforms. And then AVF central terminal to the left leg is going to give us all positive deflections when we record it on paper. So I've already covered the

20 Frontal and Horizontal Planes (revisited)

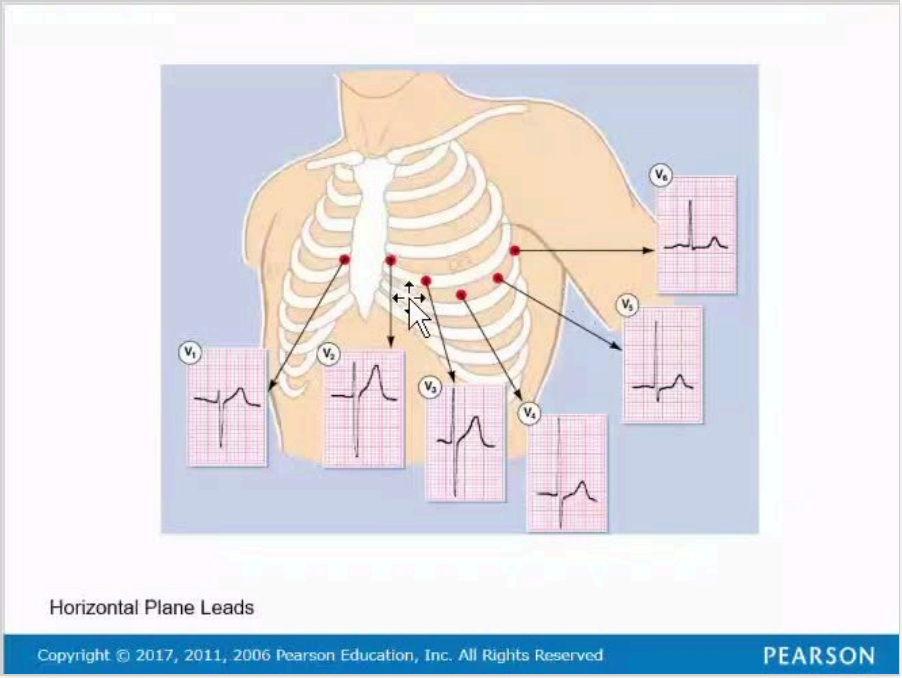
0:19:31 – 0:19:47



0:19:36 frontal and horizontal planes, just remember that the augmented leads and the pre-cordial leads are looking at the horizontal plane of the heart.

21 Horizontal Plane Leads

0:19:47 – 0:20:07

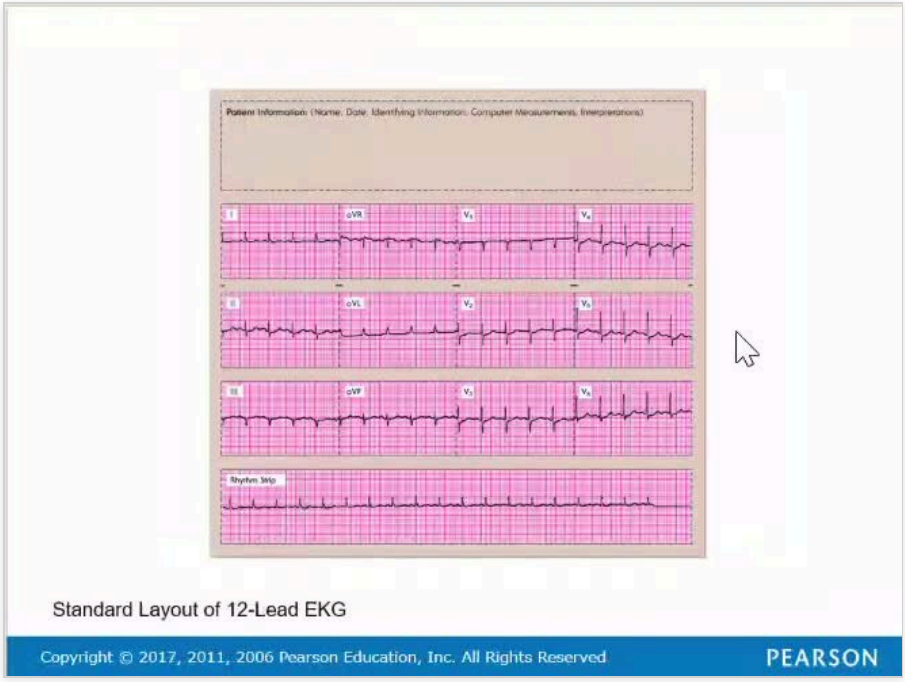


0:19:49 So then we're going to get a little snippet recording of each one of the pre-cordial leads on the chest as well. So V1, V2, V3, V4, V5, and V6.

0:20:04 And then when we put it all together on paper, this is what we're going to see.

22 Standard Layout of 12-Lead EKG

0:20:07 – 0:24:11



0:20:10 So this is a standard 12 lead EKG printout. So starting on the top right corner, we're going to get a little snippet of lead one, lead two, lead three, the augmented leads, AVR, AVL, AVF, and then we're going to get a small snippet of each one of the pre-chordial leads, V1, V2, V3, V4, V5, V6, and then at the bottom it's going to give us a running rhythm strip and that is always going to be recorded in lead 2. So why are we doing the 12 lead EKGs? Going back to

0:21:01 a person coming in with specific signs and symptoms of having a 12 lead EKG, so we want to put the 12 leads on, get a recording. So when somebody's having a myocardial infarction, it is typically caused by somebody who has coronary arteries disease or blockage in one of their coronary arteries. When they have blockage in that coronary artery, it reduces blood flow to a specific portion of the heart muscle that that blood vessel is feeding when that portion of the heart muscle doesn't get enough oxygen it can become ischemic and cause pain because those cells are not getting the oxygen level that it needs if the blockage or lack of oxygen continues that portion of the heart muscle can go from ischemia to injury to necrosis, which would be cell death. So when these types of situations happen, it disturbs the electrical flow across that section of the heart. So whatever section of the heart is affected, we will want to look at all of the leads where we see the abnormality

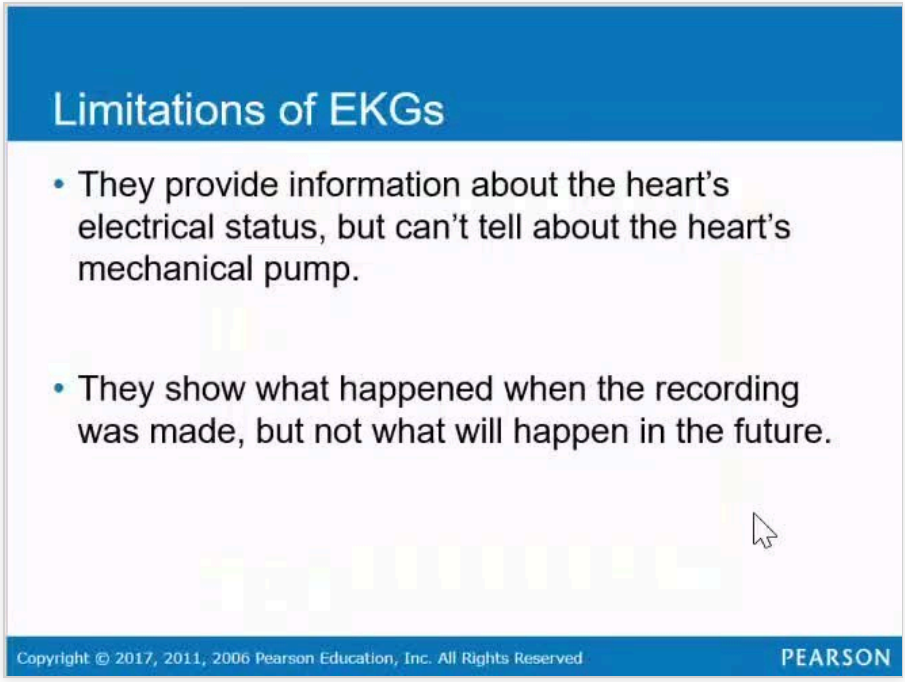
0:22:27 in the 12 lead EKG, we need to understand what lead or what surface of the heart that lead covers. So if somebody's having an MI in the front section of their heart or something that's called the anterior wall, then we would need to look at the leads that cover that anterior section

of the heart, and specifically we would be looking for things on the 12-lead EKG called ST elevations, ST depression, T-wave abnormalities, or the inversion of the T-wave. And whatever lead we see that in, we know what section of the heart that is, then we have an idea of what coronary artery actually feeds that surface of the heart. And so if we see specific abnormalities in the 12 lead EKG, this patient is going to receive fairly immediate treatment. And that would either be sent up to the cath lab, cardiac cath lab for a cardiac catheterization to be done. If the hospital doesn't have a cath lab, then maybe that patient gets a medication called TPA,

0:23:57 which can potentially dissolve the blockage inside the coronary artery, or they're going to go to the OR and have open heart surgery done. So what are the limitations

23 Limitations of EKGs

0:24:11 – 0:25:50



0:24:15 to doing 12 lead EKGs? They provide information about the heart's electrical status, but they can't tell us about the heart's mechanical pump. In order to assess the mechanical pump, we would have to feel for a pulse, perform a blood pressure, or we could put an invasive IV catheter in places where we could actually measure the heart's function. So we could put a central IV catheter in and measure specific pressures in the right atrium or the pulmonary artery or the pulmonary capillaries and also can actually measure and calculate out cardiac output. They also have limitations because they only show what happened when the recording was made. So if problems may show up when the recording is made, but typically they don't give an indication of what's going to happen in the future. So if somebody does have a heart attack and they receive treatment, typically what they will do is they'll do serial 12-lead EKGs on them.

0:25:35 So every day they'll get a 12-lead EKG so that you can track the regression or progress of the myocardial infarction. And that's going to end our lecture on 12-lead EKG.